



Provide a report on your findings from the pcap file and outline what processes / the steps you followed to achieve this. Here are each of your sub-tasks with additional instructions. Please record your findings under each sub-task title.

Sub-task 1:

- *anz-logo.jpg and bank-card.jpg are two images that show up in the users network traffic.*
- *Extract these images from the pcap file and attach them to your report.*



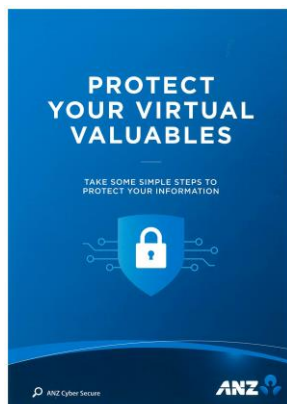
Filtered out the http responses and the image files using this command "http.response && http.content_type contains "image/jpeg".

Opened the get response "GET /anz-logo.jpg HTTP/1.1" and followed its TCP stream.

Changed the data to its raw format and did file carving by copying the range of ffd8(start of an image) to ffd9(end of an image) and then copied it to hex editor to jpeg reconstruction.

Sub-task 2:

- *The network traffic for the images "ANZ1.jpg" and "ANZ2.jpg" is more than it appears.*
- *Extract the images, include them and mention what is different about them in your report*

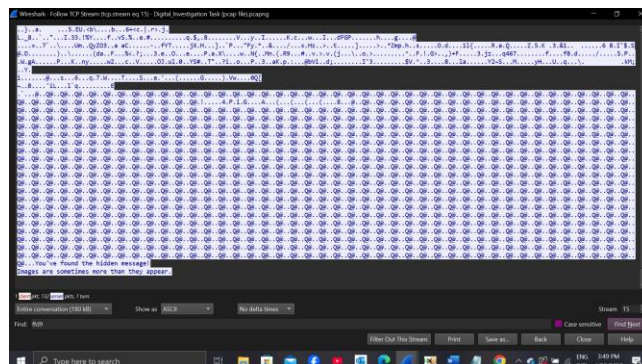


MAKE A 'PACT'

TO PROTECT YOUR VIRTUAL VALUABLES

- PAUSE before sharing your personal information
- CALL OUT suspicious messages
- ACTIVATE two layers of security with two-factor authentication
- TURN ON automatic software updates

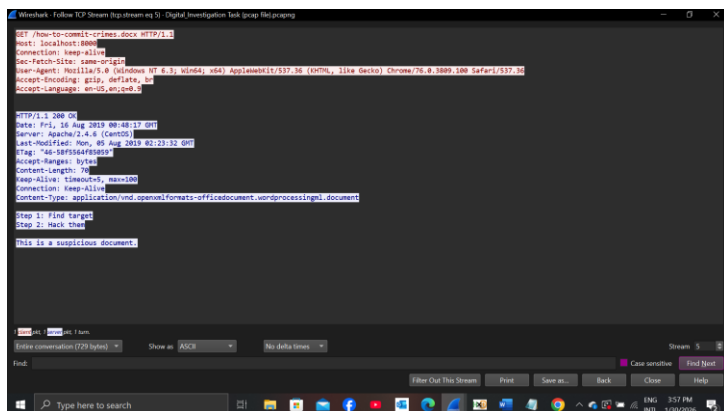
Report suspicious messages from ANZ.
Email: fraud@anz.com
Report fraudulent or unusual ANZ account activity.
137 028 +61 3 9600 7133 Corporate/Reserve Channel
133 000 +61 3 9600 8833 Personal Banking Customers



Filtered out the http responses and the imaflow the same procedure in sub task 1 but found a hidden message, when observing the raw data in ASCII formate.

Sub-task 3:

- The user downloaded a suspicious document called "how-to-commit-crimes.docx"
- Find the contents of this file and include it in your report.



```
GET /how-to-commit-crimes.docx HTTP/1.1
Host: localhost:8080
Connection: keep-alive
Sec-Fetch-Site: same-origin
User-Agent: Mozilla/5.0 (Windows NT 6.3; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/76.0.3809.100 Safari/537.36
Accept-Encoding: gzip, deflate, br
Accept-Language: en-US,en;q=0.5

HTTP/1.1 200 OK
Date: Fri, 16 Aug 2019 09:48:17 GMT
Server: Apache/2.4.6 (CentOS)
Last-Modified: Mon, 05 Aug 2019 02:23:52 GMT
ETag: "46-5f8564993935"
Accept-Ranges: bytes
Content-Length: 76
Keep-Alive: timeout=3, max=100
Connection: keep-alive
Content-Type: application/vnd.openxmlformats-officedocument.wordprocessingml.document

Step 1: Find target
Step 2: Hack them
This is a suspicious document
```

Searched for a suspicious document called “how-to-commit-crimes.docx” followed its TCP stream and viewed the content of the packet.

Sub-task 4:

- The user accessed 3 pdf documents: ANZ_Document.pdf, ANZ_Document2.pdf, evil.pdf
- Extract and view these documents. Include images of them in your report.



Searched the documents from the http responses and then since its originally a pdf file, I had to convert it to from ASCII to raw data and searched for 2550(start of a pdf) and 4446(end of a pdf) and then converted the bytes into hex using the hex editor and saved it as a pdf file.

Sub-task 5:

- The user also accessed a file called "hiddenmessage2.txt"
- What is the contents of this file? Include it in your report



Searched for "hiddenmessage2.txt", followed its TCP stream and viewed the content of the packet. Apparently, it is a JFIF file even though it was forged to look like a txt message. I file carved it by copying the range of ffd8(start of an image) to ffd9(end of an image) and then copied it to hex editor to JFIF reconstruction.

Sub-task 6:

- The user accessed an image called "atm-image.jpg"
- Identify what is different about this traffic and include everything in your report.



Searched for "atm-image.jpg", followed its TCP stream and viewed the content of the packet as usual. Converted the ASCII data to raw bytes and notice there were two sets of image (2 ranges of ffd8-ffd9) within one packet.

Sub-task 7:

- The network traffic shows that the user accessed the image "broken.png"
- Extract and include the image in your report.



Followed its TCP stream and viewed the content of the packet as usual.
Since it's a png image, I tried to find the ranges of ffd8-ffd9, but couldn't find it because this was base64 encoded.

I decoded it to Hexadecimal and reconstructed it a png file using a Hex editor.

Sub-task 8:

- The user accessed one more document called *securepdf.pdf*
- Access this document include an image of the pdf in your report. Detail the steps to access it.



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Followed its TCP stream and viewed the content of the packet as usual.
Apparently its a zip file because there was a phrase called "password is 'secure'", so I tried to find the file signature of zip file.

And then using hex editor I saved it the file in zip format and it opened me a to a pdf with two slides.